

Eigenvalues & Eigenvectors

Linear Algebra

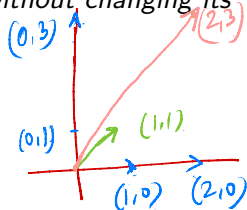
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Motivation

- When does a linear transformation stretch a vector *without changing its direction*?

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

$$A(1, 0) = (2, 0), \quad A(0, 1) = (0, 3)$$



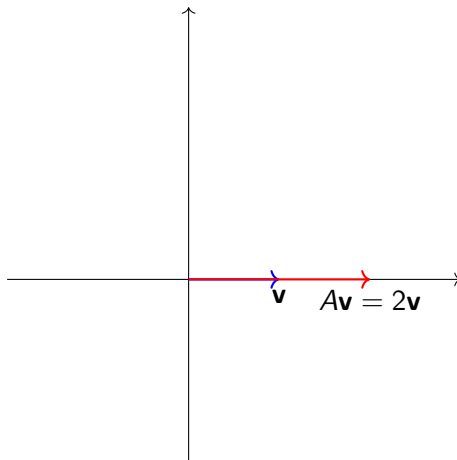
- These vectors keep their direction; only their length changes.
- Such vectors are called **Eigenvectors**.

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \neq \lambda \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Eigenvectors (informally)

Most vectors get twisted. Eigenvectors refuse to twist.

Geometric Interpretation



Direction unchanged, magnitude scaled.

Eigenpair Definition

Definition (Eigenvalue & Eigenvector)

Let $A \in M_n(\mathbb{R})$. A **NONZERO** vector $\mathbf{v}$ is called an eigenvector if

$$A\mathbf{v} = \lambda\mathbf{v}$$

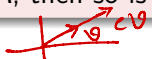
for some scalar λ . Moreover, λ is called an eigenvalue.

Remark

An eigenvector is never zero, but an eigenvalue can possibly be zero ($0 \in \mathbb{R}$). Eigenvalues and eigenvectors are also sometimes referred to as *characteristic values* and *characteristic vectors*, respectively.

Property

If $\mathbf{v}$ is an eigenvector of A , then so is $c\mathbf{v}$ for any nonzero scalar c .



Finding Eigenpairs

Finding Eigenvalues and Eigenvectors

- We have to find all those $\lambda (\in \mathbb{R})$ such that $A\mathbf{v} = \lambda\mathbf{v} \iff (A - \lambda I)\mathbf{v} = \mathbf{0}$. Now, the nontrivial solutions exist for the above system of equations if $\det(A - \lambda I) = 0$. *[Let $(A - \lambda I) \neq 0$, $(A - \lambda I)^{-1}$ exists & $\mathbf{x} = \mathbf{0}$.]*
- If $\lambda_1, \lambda_2, \dots, \lambda_k$ are k distinct eigenvalues obtained in the above step, then solve k systems of linear equations

$$(A - \lambda_i I)\mathbf{v}_i = \mathbf{0}, \text{ for } i = 1, 2, \dots, k$$

to find all the eigenvectors $\mathbf{v}_i$ corresponding to each eigenvalue.

Remark

The transformation $A - \lambda I$ becomes singular for eigenvalues. So, eigenvalues are precisely the scalars for which the transformation loses invertibility. That's profound.

$$\det(A - \lambda I) = 0 \Rightarrow A = (a_{ij})$$

$$A - \lambda I = \begin{pmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\Rightarrow \lambda^n + \left(\text{---} \right) \lambda^{n-1} + \dots + \left(\text{---} \right) \lambda^0 = 0$$

degree n-polynomial

Example

Example (Finding Eigenpairs)

Find eigenvalues of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$

Solution. First, we have

$$\det(A - \lambda I) = \begin{vmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{vmatrix}$$

By using $\det(A - \lambda I) = 0$, we deduce

$$= (4 - \lambda)(3 - \lambda) - 2$$

$$= \lambda^2 - 7\lambda + 10$$

$$(\lambda - 5)(\lambda - 2) = 0$$

Eigenvalues: $\lambda = 5, 2$. Now we will find eigenvectors corresponding to each eigenvalues.

$$\lambda_1 = 2 \quad \lambda_2 = 5$$

$$N(A - 2I_2)$$

$$\left(\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \right) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \longrightarrow \dim(\text{---}) = 1$$

||

//

$$\boxed{\lambda = 5}$$

similarly

$$N(A - 5I_2) = \text{---}$$

Characteristic Polynomial

Definition (Characteristic Polynomial)

The characteristic polynomial of $n \times n$ matrix is defined as $p_A(\lambda) = \det(A - \lambda I_n)$ where I_n is the identity matrix of size $n \times n$. The equation $p_A(\lambda) = 0$ is called the **characteristic equation**. Note that this equation is in terms of λ .

Properties:

- Degree of $p_A(\lambda) = n$
- Sum of eigenvalues = $\text{tr}(A)$
- Product of eigenvalues = $\det(A)$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} \rightarrow 2 \times 2$$

$$\begin{aligned} &= (a-\lambda)(d-\lambda) - bc \\ &= ad - (a+d)\lambda + \lambda^2 - bc \\ &= \lambda^2 - (a+d)\lambda + (ad-bc) \end{aligned}$$

Some Important Results

Proposition

An $n \times n$ matrix A is invertible if and only if 0 is not an eigenvalue.

Theorem

Let λ and $\mathbf{v}$ be an eigenpair for the $n \times n$ matrix A . If k is an arbitrary real number, then $\mathbf{v}$ is also an eigenvector of the matrix $A - kI$ corresponding to the eigenvalue $\lambda - k$.

From the hyp. $\Rightarrow A\mathbf{v} = \lambda\mathbf{v} \Rightarrow A\mathbf{v} - k\mathbf{v} = \lambda\mathbf{v} - k\mathbf{v}$

$$\begin{aligned} (A - kI)\mathbf{v} &= \lambda\mathbf{v} - k\mathbf{v} \\ &= (\lambda - k)\mathbf{v} \end{aligned} \quad \left| \quad \begin{aligned} &\Rightarrow \underbrace{(A - \lambda I)}_B \mathbf{v} = (\lambda - k)\mathbf{v} \\ &\Rightarrow \boxed{B\mathbf{v} = (\lambda - k)\mathbf{v}} \end{aligned} \right.$$

More on Characteristic Polynomial

Definition (Algebraic Multiplicity of an Eigenvalue)

Let A be an $n \times n$ matrix with characteristic polynomial given by $p_A(\lambda)$. Then, the multiplicity of an eigenvalue of $p_A(\lambda)$ is the number of times λ occurs as a root of that characteristic polynomial. The number of occurrences is called the algebraic multiplicity of the said eigenvalue.

$$(\lambda - 1)^2 (\lambda - 2) = 0$$

$$\lambda = \underbrace{1, 1}_2, 2$$

1 - has algebraic multiplicity of 2
2 - has algebraic multiplicity of order 1

More on Characteristic Polynomial. Contd.

- For a given $n \times n$ matrix $A = (a_{ij})$, the characteristic polynomial $p_A(\lambda)$ assumes the form

$$p_A(\lambda) = \det(A - \lambda I) = \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix}$$

- Upon expanding, we get a polynomial in λ with leading coefficient $(-1)^n$ and we get

$$p_A(\lambda) = (-1)^n \lambda^n + b_1 \lambda^{n-1} + b_2 \lambda^{n-2} + \cdots + b_n,$$

where $b_1, b_2, \dots, b_n \in \mathbb{R}$.

More on Characteristic Polynomial. Contd.

- Although the above polynomial may not have n real roots. But if we consider the roots in *Complex Numbers*, the above polynomial will have n roots (not necessarily distinct), thanks to the *Fundamental Theorem of Algebra*. If we let $\lambda_1, \lambda_2, \dots, \lambda_k$ denote the distinct eigenvalues of A , then $p_A(\lambda)$ can be factored as

$$p_A(\lambda) = (-1)^n (\lambda - \lambda_1)^{m_1} (\lambda - \lambda_2)^{m_2} (\lambda - \lambda_3)^{m_3} \dots (\lambda - \lambda_k)^{m_k},$$

where since $p_A(\lambda)$ has degree n

$$m_1 + m_2 + \dots + m_n = n.$$

Definition (Eigenspace)

Let A be an $n \times n$ matrix. For a given eigenvalue λ_i , let E_i denote the set of all vectors satisfying $A\mathbf{v} = \lambda_i\mathbf{v}$. Then, E_i is called the eigenspace of A corresponding to the eigenvalue λ_i . Thus, E_i is the solution to the linear system $(A - \lambda_i I)\mathbf{v} = \mathbf{0}$.

$$E_i \rightarrow N(A - \lambda_i I) \leftarrow \lambda_i$$

Algebraic vs Geometric Multiplicity

Definition (Geometric Multiplicity)

The dimension of the eigenspace:

$$\ker(A - \lambda I)$$

is called the Geometric Multiplicity of A with respect to λ .

Remark

In general

$$A \in M_n(\mathbb{R})$$

$$1 \leq \text{Geometric Multiplicity} \leq \text{Algebraic Multiplicity}$$

Definition (Defective Eigenvalue)

When the geometric multiplicity of a repeated eigenvalue is strictly less than its algebraic multiplicity, then that eigenvalue is said to be *defective*.

Theorem

Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Proof. • Suppose $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly dependent. Let k be the smallest positive integer such that $\mathbf{v}_k \in \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_{k-1}\}$.

• Thus there exist $a_1, \dots, a_{k-1} \in \mathbf{R}$ such that

$$\mathbf{v}_k = a_1\mathbf{v}_1 + \dots + a_{k-1}\mathbf{v}_{k-1}. \quad (1)$$

• Apply A to both sides of this equation, getting

$$\lambda_k\mathbf{v}_k = (a_1 - \lambda_1)\mathbf{v}_1 + \dots + (a_{k-1} - \lambda_{k-1})\mathbf{v}_{k-1}. \quad (2)$$

• Multiply both sides of (1) by λ_k and then subtract from (2), we get

$$\mathbf{0} = a_1(\lambda_k - \lambda_1)\mathbf{v}_1 + \dots + a_{k-1}(\lambda_k - \lambda_{k-1})\mathbf{v}_{k-1}. \quad (3)$$

- Because we chose k to be the smallest positive integer satisfying (1), the set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_{k-1}\}$ is linearly independent.
- Thus the equation above implies that all a 's are 0 (recall that λ_k is not equal to any of $\lambda_1, \dots, \lambda_{k-1}$).
- However, this means that $\mathbf{v}_k$ equals 0 (see (2)), contradicting our hypothesis that all $\mathbf{v}$'s are nonzero.
- Therefore, our assumption that $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly dependent must have been false. □

Example

Example (Finding Eigenspace)

Determine all eigenspaces for the matrix $A = \begin{pmatrix} 5 & -4 \\ 8 & -7 \end{pmatrix}$. ✓

Matrix Polynomial

Definition (Matrix polynomial)

Let $p : \mathbb{R} \rightarrow \mathbb{R}$ is an ordinary polynomial

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

then, we can use p to define, by extension, an analogous matrix polynomial

$$p(A) = a_0I + a_1A + a_2A^2 + \cdots + a_nA^n$$

provided that the entries of the square matrix A belong to $\mathbb{R}$.

Example (Matrix Polynomial)

Let $A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$. Find the matrix polynomial $p(A) = 5I - 5A + A^2$. (279)

(2x2) $\begin{pmatrix} & \end{pmatrix} = p(A) = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} - \begin{pmatrix} 15 & 5 \\ 5 & 10 \end{pmatrix} + \begin{pmatrix} 31 & 31 \\ 12 & 12 \end{pmatrix}$

Cayley–Hamilton Theorem

Theorem

For an $n \times n$ matrix A , the characteristic polynomial $p_A(\lambda) = \det(A - \lambda I)$ is given as

$$p_A(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_0,$$

then $p_A(A) = 0$. That is,

$\in \mathbb{R}$ $A^n + a_{n-1}A^{n-1} + \cdots + a_0I = 0.$ $0_{n \times n}$

Remark

- In other words, when you replace the matrix A with itself in its characteristic polynomial, the outcome is consistently the zero matrix.
- The eigenvalues of the matrix are represented by the roots of this polynomial.

Verification Example

Example (Verification of C-H Theorem)

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Check for C-H theorem.

Characteristic polynomial:

$$\lambda^2 - 4\lambda + 3$$

Cayley-Hamilton says:

$$A^2 - 4A + 3I = 0$$

$n \times n$

Application: Finding Inverse

- For any 2×2 matrix A , we have

$$A^2 - (\operatorname{tr} A)A + (\det A)I = 0$$

- If $\det(A) \neq 0$,

$$A^{-1} = \frac{1}{\det A}[(\operatorname{tr} A)I - A].$$

Finding Inverse of A

Theorem (Inverse via Cayley–Hamilton)

Let $A \in M_n(\mathbb{R})$ be an invertible matrix with characteristic polynomial

$$p(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0.$$

Then

$$A^{-1} = \boxed{\frac{-1}{a_0}} \left(A^{n-1} + a_{n-1}A^{n-2} + \cdots + a_1I \right).$$

$$a_0 = \det(A)$$

In particular, A^{-1} is a polynomial in A of degree at most $n - 1$.

A^{-1} exists $\Rightarrow a_0 \neq 0$?

$$\boxed{a_0 = 0}$$

$$p(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda$$

$$p(\lambda) = 0, \lambda = 0 \Rightarrow \det(A) = 0 \Rightarrow \Leftarrow$$

Proof of Theorem

Proof.

By the Cayley–Hamilton theorem, A satisfies its own characteristic equation:

$$A^n + a_{n-1}A^{n-1} + \cdots + a_1A = -a_0I.$$

Factor A from the left-hand side:

$$A\left(A^{n-1} + a_{n-1}A^{n-2} + \cdots + a_1I\right) = -a_0I.$$

Since A is invertible, $\det(A) \neq 0$, and hence $a_0 = (-1)^n \det(A) \neq 0$. Therefore,

$$A\left(\frac{-1}{a_0}\left(A^{n-1} + a_{n-1}A^{n-2} + \cdots + a_1I\right)\right) = I,$$

which proves the expression. □

Application: Computing Powers

- If we are given

$$\boxed{p_A(\lambda) = \lambda^2 - 4\lambda + 3} \rightarrow \text{Since } A \rightarrow 2 \times 2$$

$$A^2 = 4A - 3I$$

Then any power A^n can be written as:

$$A^n = \alpha A + \beta I$$

Reduces large powers to simple form.

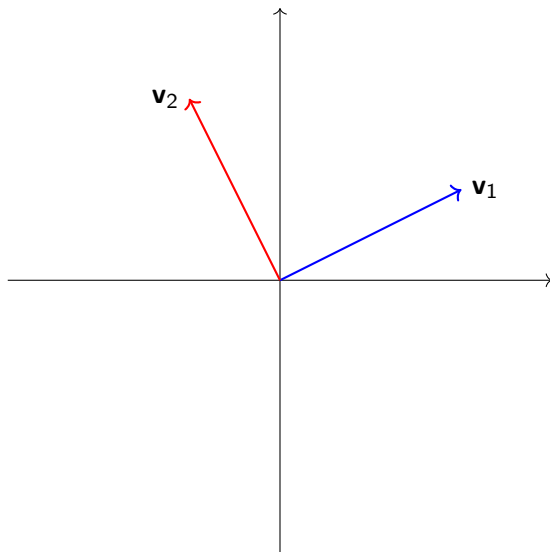
$$\boxed{A^3}$$

$$A^3 = 4A^2 - 3A$$

$$= 4(4A - 3I) - 3A$$

$$= 13A - 12I$$

Geometric Picture of Two Eigenvectors



Transformation stretches along eigen-directions.

Summary

- $A\mathbf{v} = \lambda\mathbf{v}$
- $\det(A - \lambda I) = 0$
- Characteristic polynomial
- Algebraic vs Geometric multiplicity
- Diagonalization
- Cayley–Hamilton theorem
- Applications to inverse and powers

Exercises

- 1 Find eigenvalues of $\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$.
- 2 Verify Cayley–Hamilton for $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.
- 3 Use CH theorem to compute A^5 for a 2×2 matrix.

$$A = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}$$

$$A^2 = \begin{pmatrix} a_{11}^2 & 0 & \dots & 0 \\ 0 & a_{22}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn}^2 \end{pmatrix}$$

$$\boxed{\lambda_i = a_{ii} \quad i = 1(1)n}$$

$$a_{ii} \neq 0, \quad i = 1(1)n.$$

A^{-1} exists and

$$A^{-1} = \begin{pmatrix} a_{11}^{-1} & 0 & \dots & 0 \\ 0 & a_{22}^{-1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn}^{-1} \end{pmatrix}$$

$$\boxed{a_{ii} \neq 0, \quad i = 1(1)n}$$

$$\begin{aligned} P^{-1}AP &= B \quad (P^{-1} \text{ exists}) \\ B^2 &= BB = (P^{-1}AP)(P^{-1}AP) \\ &= P^{-1}A(PP^{-1})AP = P^{-1}A^2P \end{aligned}$$

Diagonalization Idea

Definition (Similarity)

Suppose A is an $n \times n$ matrix. We say that an $n \times n$ matrix B is similar to A if there is an invertible $n \times n$ matrix P so that

$$B = P^{-1}AP.$$

$$\begin{aligned} PB &= AP \\ \Rightarrow PBP^{-1} &= A \quad (\mathcal{B}^{-1} = P) \\ \Rightarrow (\mathcal{B}^{-1})^{-1} B \mathcal{B} &= A \end{aligned}$$

Remark

Let $T : V \rightarrow V$ be a linear transformation where V is a non-zero finite dimensional vector space. If A is the matrix of T with respect to a basis $\mathcal{A}$ and if B is the matrix with respect to another basis $\mathcal{B}$ of V , then $B = P^{-1}AP$ where P is the change of basis matrix from the basis $\mathcal{A}$ to $\mathcal{B}$.

Definition (Diagonalizability)

A square matrix is said to be *diagonalizable* if it is similar to a diagonal matrix.

$A \rightarrow$ Diagonalizable.

if we can find out P s.t P^{-1} exists and

$$P^{-1} \underbrace{\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}}_A P = \underbrace{\begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{pmatrix}}_{B = \text{diag}(d_1, d_2, \dots, d_n)}$$

Diagonalizability

$$A = P^{-1}AP \quad \text{---} \quad Av = \lambda v \quad | \quad Bv_1 = \lambda v_1$$

Theorem (Eigenvalues of Similar Matrices)

Similar matrices have the same eigenvalues (including multiplicities).

Theorem

An $n \times n$ matrix A is similar to a diagonal matrix if and only if A is nondefective. In such a case, if $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ denote n linearly independent eigenvectors of A and

$$S = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n],$$

then

$$S^{-1}AS = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n),$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigenvalues of A (not necessarily distinct) corresponding to the eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$.

*(S^{-1} exists
columns are
independent)*

Diagonalizable Matrices: Counter Examples

1. is a defective eigen value!

Example (Invertible but NOT Diagonalizable)

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then A is invertible but not diagonalizable.

$$\det(A - \lambda I) = 0 \Rightarrow \det \begin{pmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{pmatrix} = 0$$

$\boxed{\lambda = 1}$ $\rightarrow$ algebraic multiplicity of 2.

$$N(A - I) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad | \quad x_2 = 0$$

$\left(1, \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) \leftarrow$ eigen pair. $\dim$ Eigen space $(\lambda=1) = 1$ $\boxed{G.M. = 1}$

Diagonalizable Matrices: Counter Examples

Diagonalizability and Invertibility are Not Correlated

Among the following matrices, the first is diagonalizable and invertible, the second is diagonalizable but not invertible, the third is invertible but not diagonalizable, and the fourth is neither invertible nor diagonalizable, as the reader can verify:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Note/Remark on Diagonalizability

Remark

An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Note

If A has n linearly independent eigenvectors:

$$A = PDP^{-1}$$

where D is diagonal with eigenvalues.

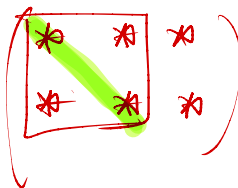
Then:

$$A^k = PD^kP^{-1}$$

Thank You



3×2 .



2×3

Questions?